

EDUCATION

- **Instituto Superior Técnico - University of Lisbon** Lisbon, Portugal
PhD in Electrical and Computer Engineering; *Sept. 2023 – Present*
- **Instituto Superior Técnico - University of Lisbon** Lisbon, Portugal
Master's degree in Aerospace Engineering; *Sept. 2018 – 2020*
- **Indian Institute of Technology Madras** Chennai, India
Integrated Master in Aerospace Engineering; *Aug. 2018 – Nov. 2018*
- **Instituto Superior Técnico - University of Lisbon** Lisbon, Portugal
Bachelor's degree in Aerospace Engineering; *Sept. 2015 – July 2018*
- **Centro de Estudos de Fátima** Fátima, Portugal
High School - Science and Technology; *Sept. 2012 – Jun. 2015*

EXPERIENCE

- **Instituto de Telecomunicações** Lisbon, Portugal
ML/NLP researcher *Sept. 2023 - Present*
 - **ML/NLP researcher:** Research on structured memories and attention mechanism for natural language generation in the scope of DECOLLAGE project under the supervision of professor André Martins.
- **NTT DATA Europe & Latam** Lisbon, Portugal
Assistant Engineer *Sept. 2022 - Sept. 2023*
 - **Data scientist in optical character recognition project:** End-to-end solution whose purpose refers to the automation of the account opening process for new customers, through the automatic reading of digitized documents and subsequent validation of the information not only between different types of documents but also between documents and a local repository of the client.
- **Institute For Systems and Robotics** Lisbon, Portugal
Master's Thesis Intern; Grade: 19/20 *Jul. 2020 - May. 2022*
 - **System identification of mechanical systems using deep learning methods:** Development of physical-aware deep learning architectures for system identification of mechanical systems. The novelty resides on combining discrete variational principles with state-of-the-art deep learning techniques by means of implicit differentiation.
- **Albatroz Engenharia** Lisbon, Portugal
Internship *Jul. 2019 - Sept. 2019*
 - **Hardware:** Implementation of an open glider network node in Castelo Branco aerodrome: construction and programming of a receiver for FLARM/OGN tracker equipped aircraft.
 - **Software:** Development of a support airfield web application for collision prevention and support in aircraft landing based on the receiver installed.

CERTIFICATIONS

- **Python for Data Analysis & Visualization 2022** *2022*
Udemy
- **Python A – ZTM: Python For Data Science With Real Exercises!** *2022*
Udemy
- **Microsoft Certified: Azure Fundamentals** *2022*
Microsoft
- **Fundamentals of Reinforcement Learning** *2020*
Online non-credit course authorized by University of Alberta and offered through Coursera

Machine Learning

Online non-credit course authorized by Stanford University and offered through Coursera

2020

Certificate in Drone Simulation and Control

Instituto Superior Técnico

2018

PUBLICATIONS

- **Saul Santos**, M. Ekal, and R. Ventura. Symplectic momentum neural networks - using discrete variational mechanics as a prior in deep learning. In *Proceedings of The 4th Annual Learning for Dynamics and Control Conference*, volume 168 of *Proceedings of Machine Learning Research*, pages 584–595.
- **Saul Santos**, Vlad Niculae, Daniel C McNamee, and Andre F.T. Martins. Sparse and structured hopfield networks. In International Conference on Machine Learning, 2024b,
- **Saul Santos**, Vlad Niculae, Daniel McNamee, and Andre F. T. Martins. Hopfield-fenchel-young networks: A unified framework for associative memory retrieval. arXiv preprint arXiv:2411.08590, 2024a. <https://arxiv.org/abs/2411.08590>.
- **Saul Santos**, António Farinhas, Daniel C McNamee, and Andre F.T. Martins. ∞ -Video: A Training-Free Approach to Long Video Understanding via Continuous-Time Memory Consolidation. In International Conference on Machine Learning, 2025b,
- **Saul Santos**, António Farinhas, Daniel C. McNamee, and André Martins. Modern Hopfield Networks with Continuous-Time Memories. In *New Frontiers in Associative Memories*, 2025. <https://openreview.net/forum?id=bU4dyLTnp3>
- Emmanouil Zaranis*, António Farinhas*, **Saul Santos***, Beatriz Canaverde*, Miguel Moura Ramos*, Aditya K. Surikuchi, André Viveiros, Baohao Liao, Elena Bueno-Benito, Nithin Sivakumaran, Pavlo Vasylenko, Shoubin Yu, Sonal Sannigrahi, Wafaa Mohammed, Ben Peters, Danae Sánchez Villegas, Elias Stengel-Eskin, Giuseppe Attanasio, Jaehong Yoon, Stella Frank, Alessandro Suglia, Chrysoula Zerva, Desmond Elliott, Mariella Dimiccoli, Mohit Bansal, Oswald Lanz, Raffaella Bernardi, Raquel Fernández, Sandro Pezzelle, Vlad Niculae, and André F. T. Martins. Movie Facts and Fibs (MF²): A Benchmark for Long Movie Understanding. arXiv preprint arXiv:2506.06275, 2025. <https://arxiv.org/abs/2506.06275>

* means equal contribution

SKILLS AND INTERESTS

- **Languages:** C, Python, Matlab, Javascript.
- **Technologies:** MS office tools (Excel, Word, Powerpoint), Sony Vegas, Adobe Photoshop, Solidworks, Autodesk Inventor, Solidedge, Latex
- **Libraries:** PyTorch, Seaborn, Matplotlib, NumPy, Pandas, Scikit-Learn, SciPy.
- **Technical Skills:** Microsoft Azure.
- **Soft Skills:** Critical Thinking, Problem Solving.
- **Areas of Interest:** Avionics, Aeronautics, Space, Control, Electronics, Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing.
- **Other Interests:** Cinema, space, music, technology, sports, photography, nature, traveling.

LANGUAGES

Portuguese (mother tongue), English (fluent), French (elementary).